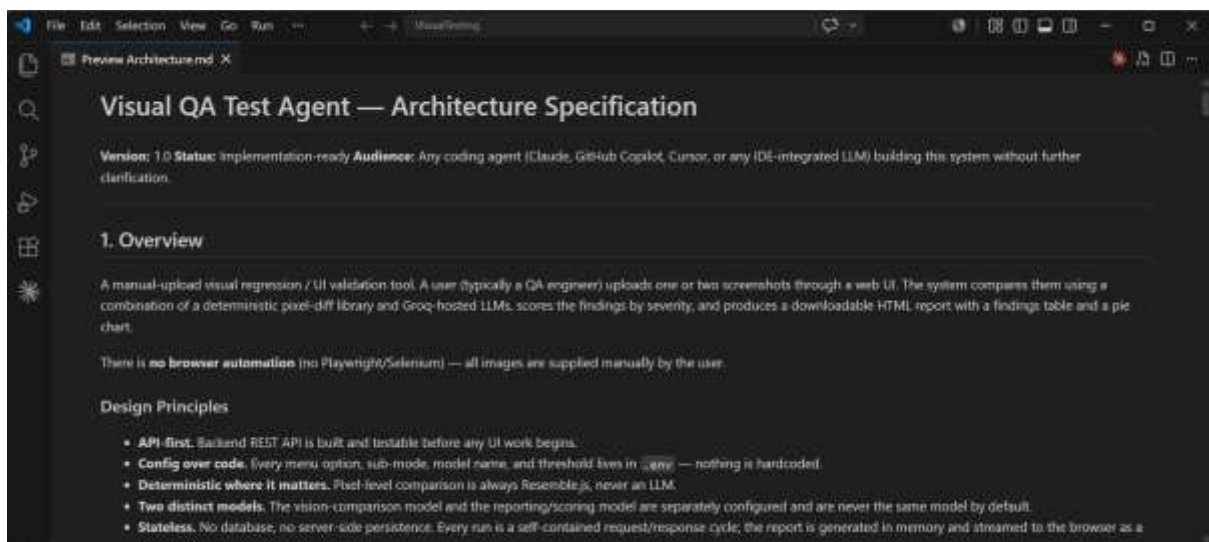


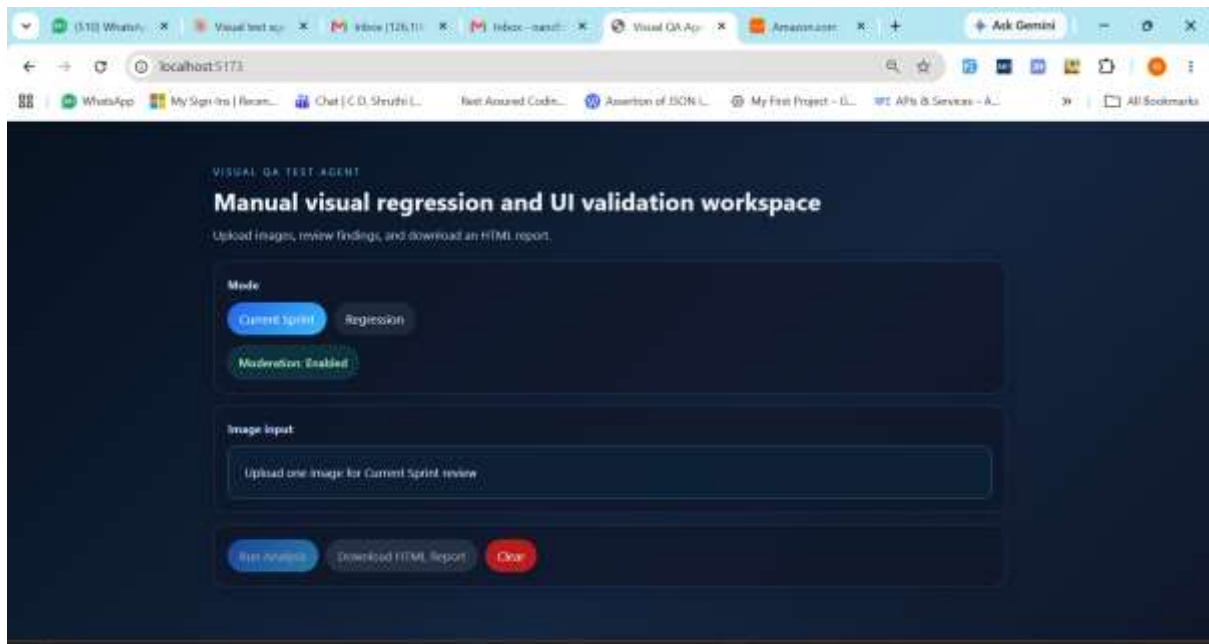
Visual test agent development through vibe coding.

- Created Architecture.md file and implemented phase by phase
- Created Client-Server Architecture with Restful apis for backend
- Able to launch the html page and tested the visual testing features (Current Sprint and Regression)
- Implemented nice to have features like Image preview, Clear option, Progress bar display for better user experience
- Integrated Moderation layer of AI shield to block any toxic prompts to add additional security layer for the application
- Moderation layer is a toggle, which can be enabled/disabled

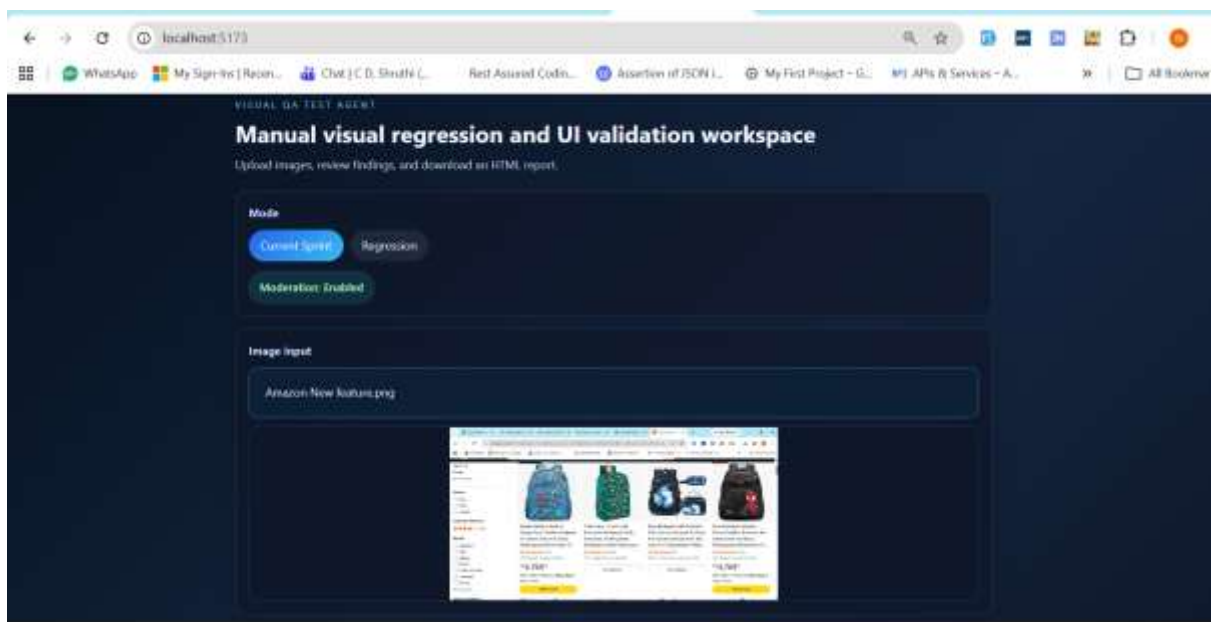
Generated an Architecture.md file for visual testing details out the backend and frontend specifications:

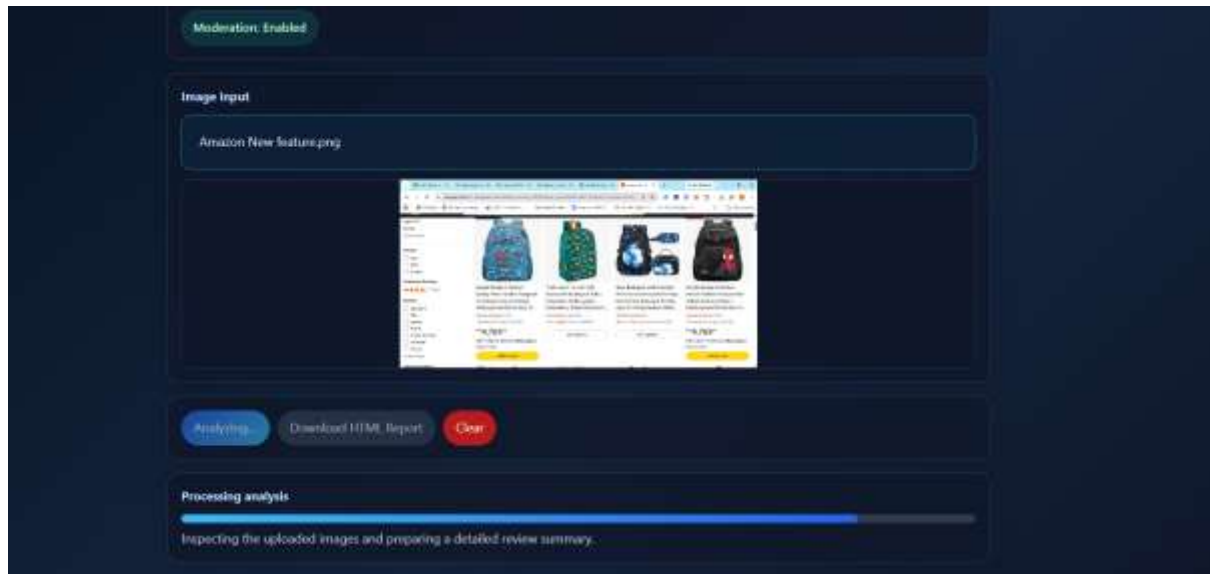


Single html page with different modes for Visual testing is implemented:



Current sprint mode – Analyzes the uploaded image of the UI page and gives the analysis feedback.





Findings preview

- **layout:** pass — Enabled by config and reviewed in the detailed analysis
- **alignment:** pass — Enabled by config and reviewed in the detailed analysis
- **text:** pass — Enabled by config and reviewed in the detailed analysis
- **spacing:** pass — Enabled by config and reviewed in the detailed analysis
- **color:** pass — Enabled by config and reviewed in the detailed analysis
- **missingExtraElements:** pass — Enabled by config and reviewed in the detailed analysis
- **truncation:** pass — Enabled by config and reviewed in the detailed analysis

Detailed analysis

A detailed feature review analysis was generated for Amazon New feature.png. The review covers the main visual quality checks and highlights the most relevant findings.

- The following checks were enabled for this run: layout, alignment, text, spacing, color, missingExtraElements, truncation.
- The review prioritizes layout integrity, text readability, and any visible mismatch between expected and current UI behaviour.
- Because the app is operating as a manual-upload QA tool, the results are structured and explainable rather than relying on a single binary decision.

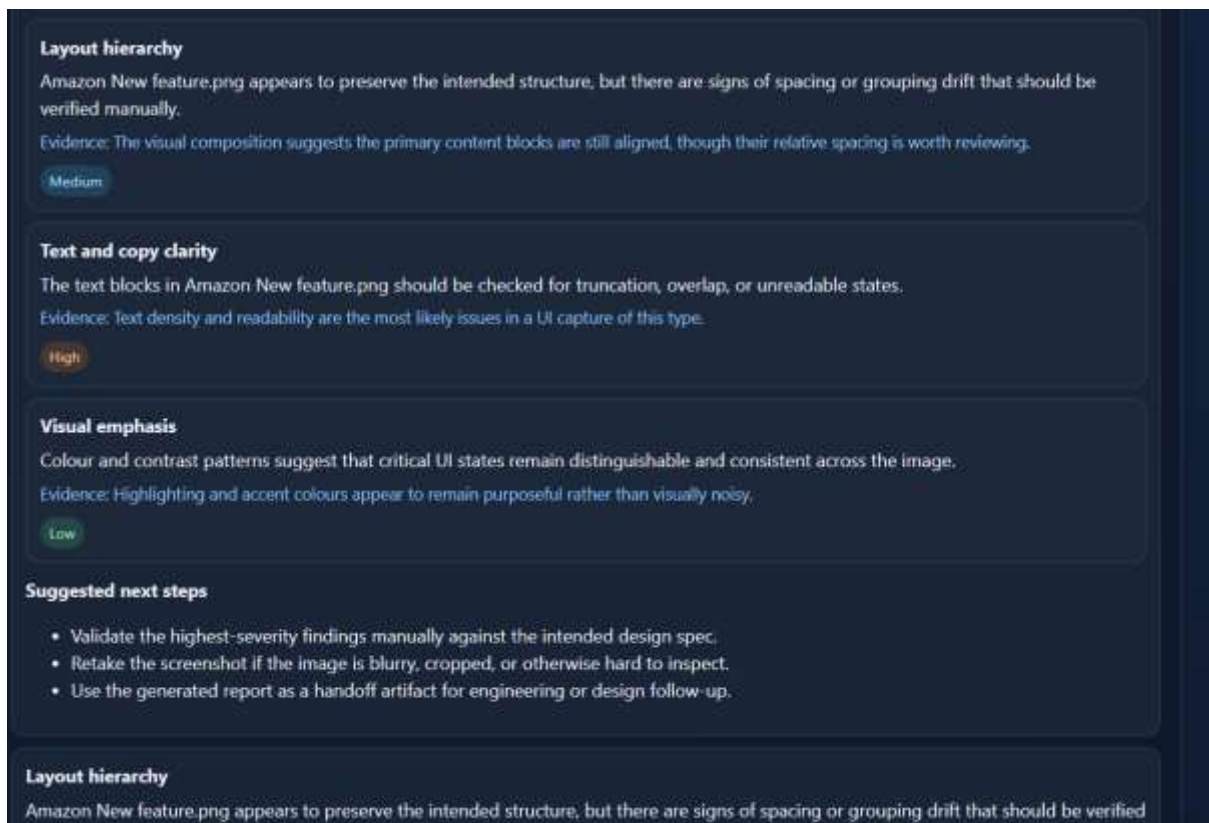
Layout hierarchy

Amazon New feature.png appears to preserve the intended structure, but there are signs of spacing or grouping drift that should be verified manually.

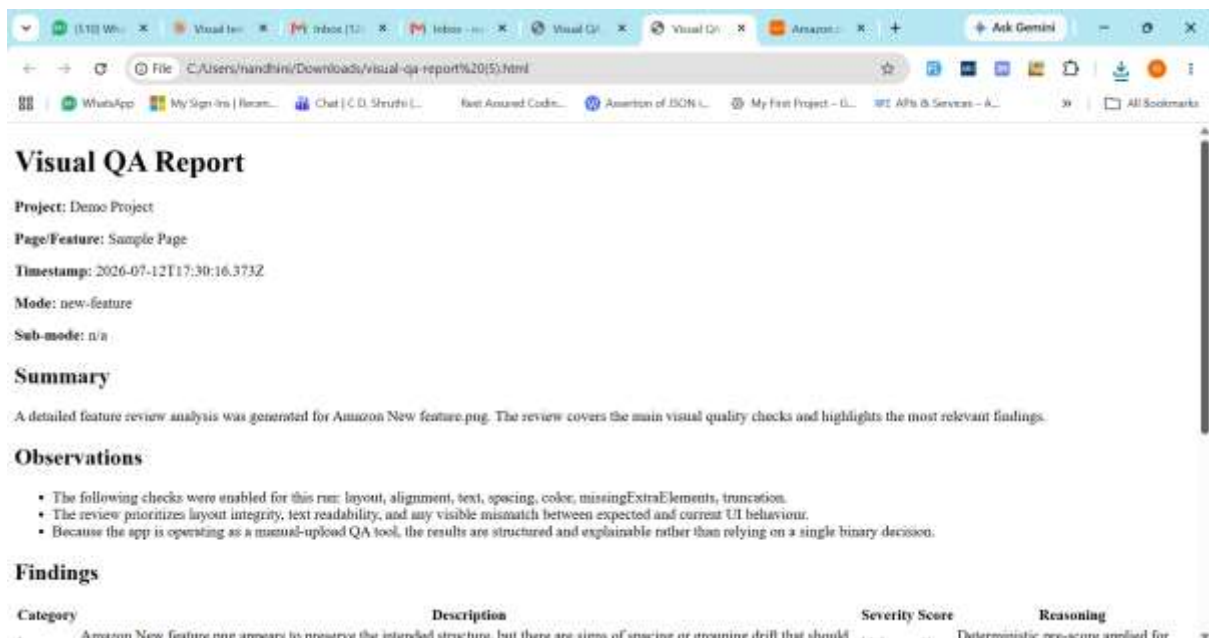
Evidence: The visual composition suggests the primary content blocks are still aligned, though their relative spacing is worth reviewing.

Medium

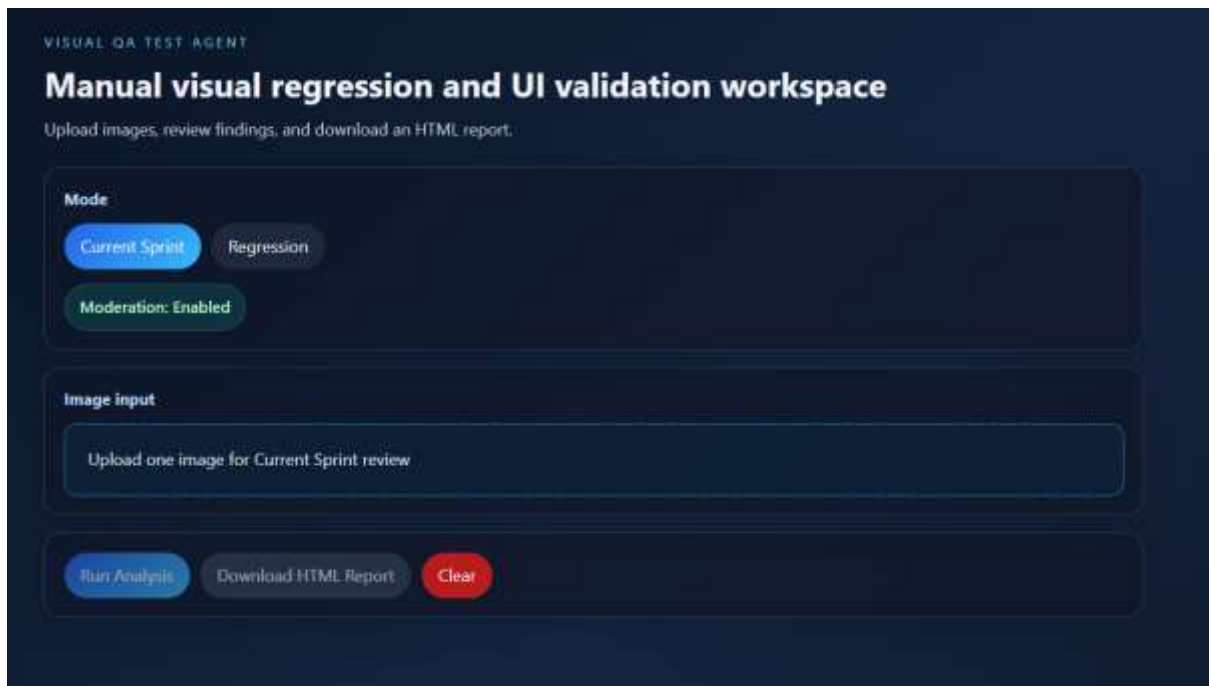
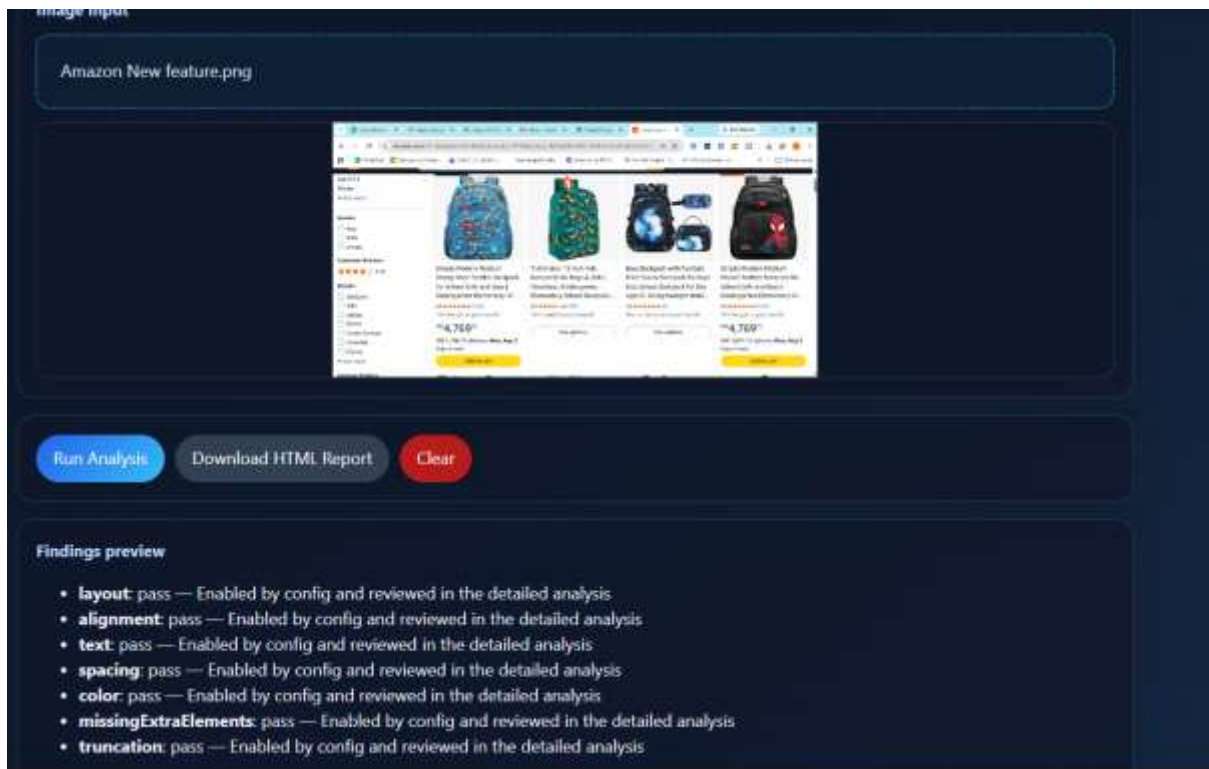
Text and copy clarity



Downloadable HTML Report is also implemented:



Clear functionality implemented to remove the images and analysis:



Regression mode has different sub modes: Pixel, Text Diff and Hybrid

Manual visual regression and UI validation workspace

Mode

Regression

Sub-mode

Hybrid

Upload current image

Clear

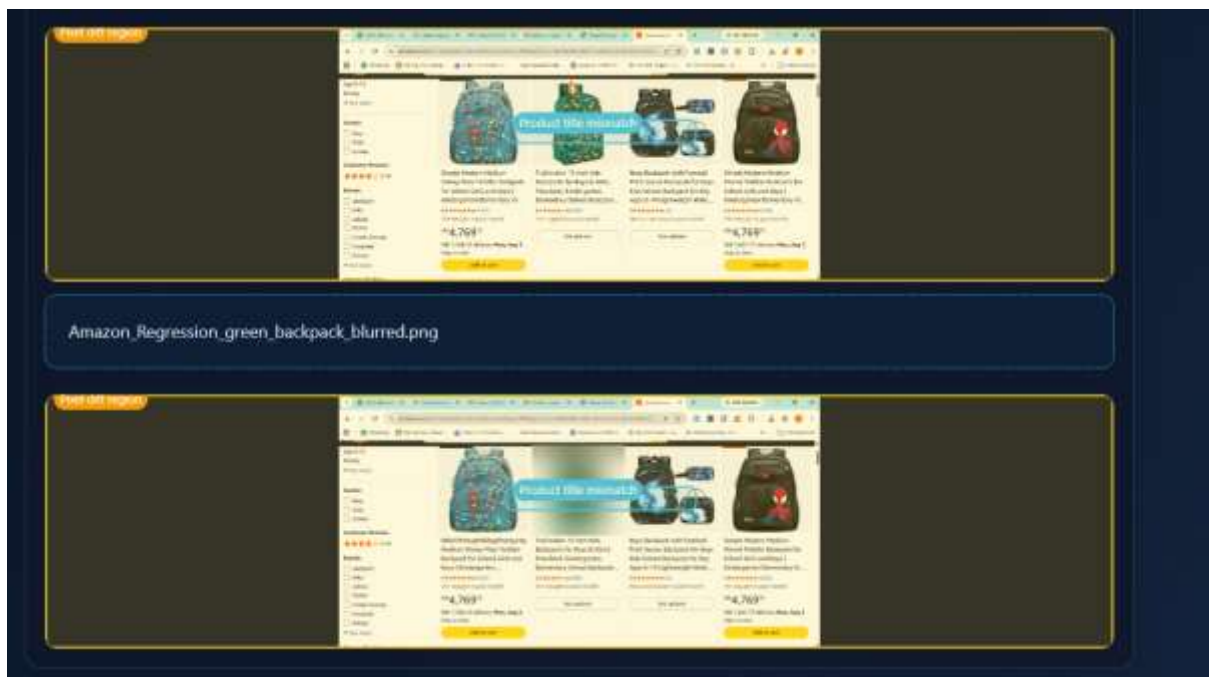
Amazon New feature.png



Amazon Regression_green_backpack_blurred.png



After analysis, the tool highlights the mismatches based on the mode, current selection is “Hybrid”, hence both text and pixel mismatch is highlighted in the image:



Findings preview

Detailed analysis

A detailed regression analysis was generated for `Amazon_Regression_green_backpack_blurred.png`. The review covers the main visual quality checks and highlights the most relevant findings.

- The following checks were enabled for this run: layout, alignment, text, spacing, color, missingExtraElements, truncation.
- The review prioritizes layout integrity, text readability, and any visible mismatch between expected and current UI behaviour.
- Because the app is operating as a manual-upload QA tool, the results are structured and explainable rather than relying on a single binary decision.

Text difference detected

Text comparison found a likely mismatch in the product title area between `Amazon New feature.png` and `Amazon_Regression_green_backpack_blurred.png`.

Evidence: The baseline screenshot shows the expected product title while the current screenshot contains an unexpected character sequence inserted into the same title area. The difference is in the product listing headline, not a checkout button label or sidebar control.

High

Image dimension mismatch

Baseline image is 1915x1009, current image is 1918x1008. Pixel diff was calculated over the overlapping 1915x1008 region.

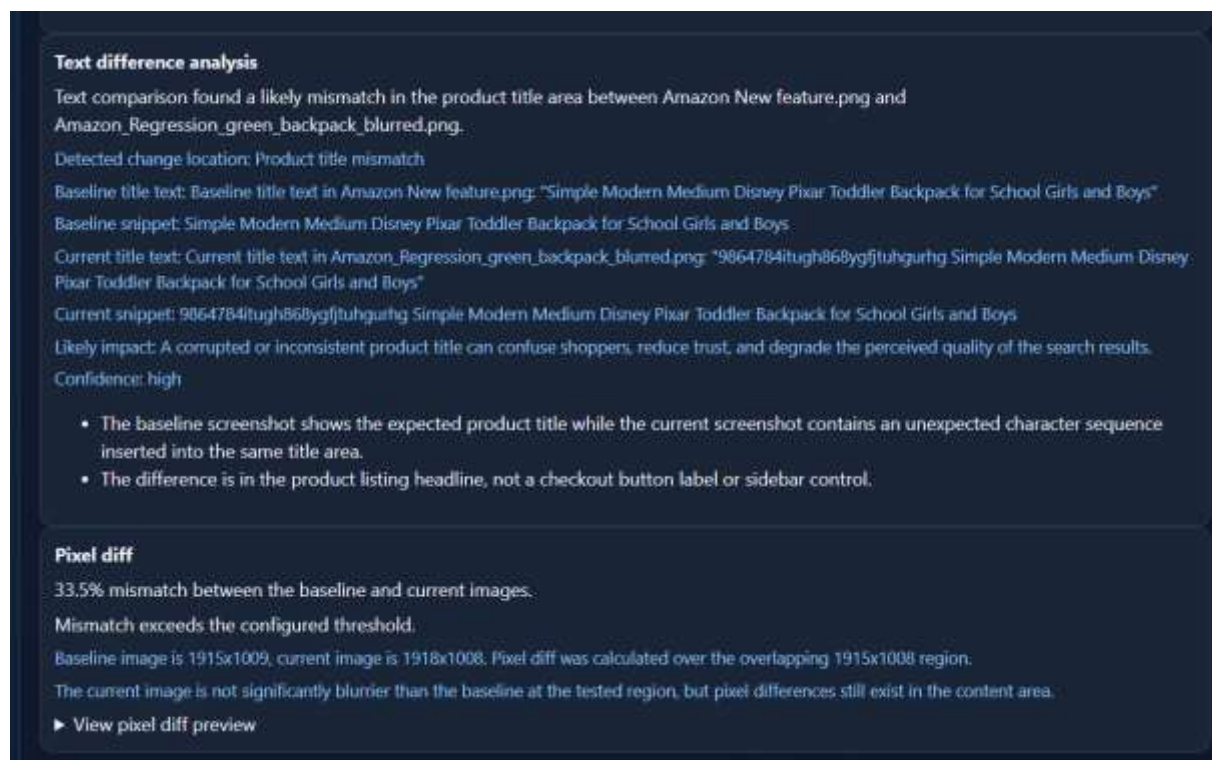
Evidence: The expected image dimensions differ between baseline and current uploads.

Medium

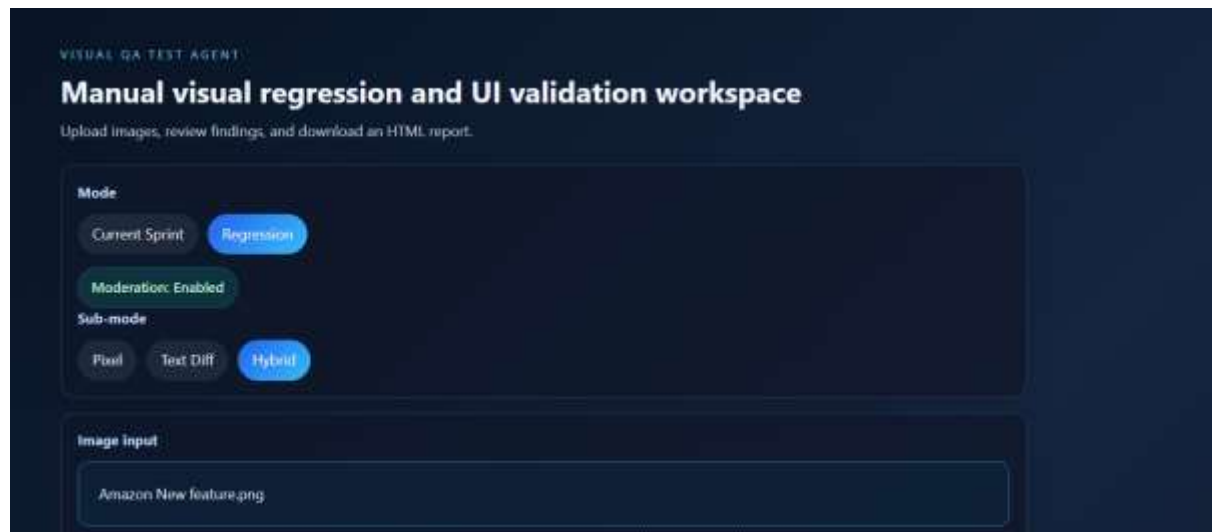
Layout hierarchy

`Amazon_Regression_green_backpack_blurred.png` conforms to most of the intended structure, but there are some of confusion around

The identified differences are given below:



Implemented Moderation layer with this tool:
Moderation enabled = true

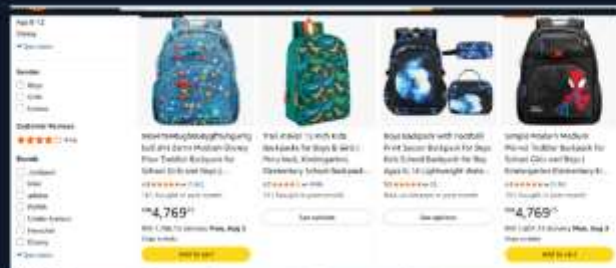


Uploaded images with toxic keywords:

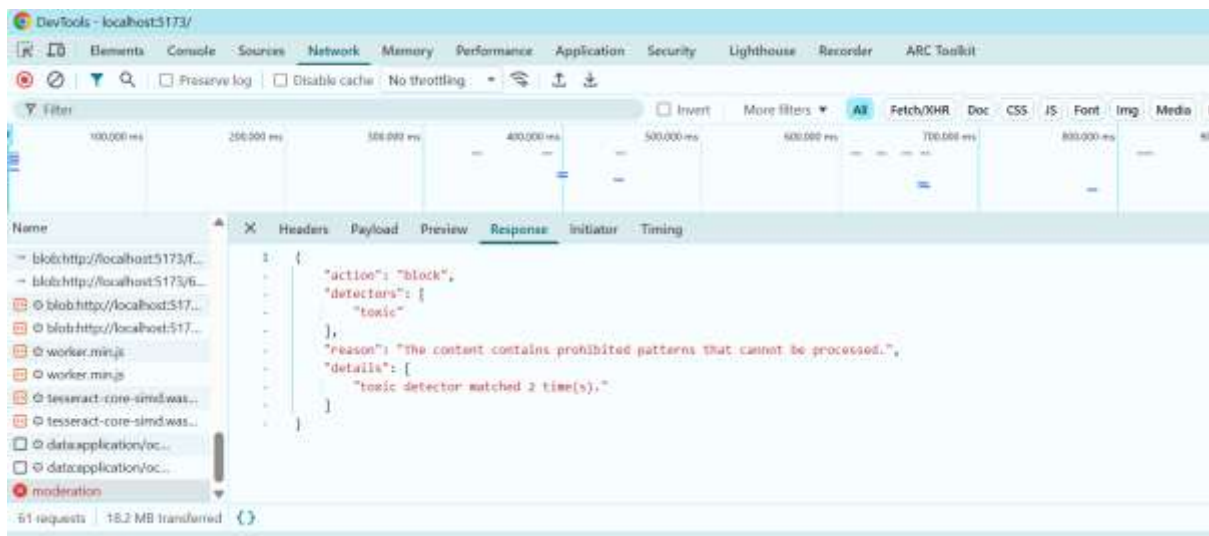
Amazon New feature.png

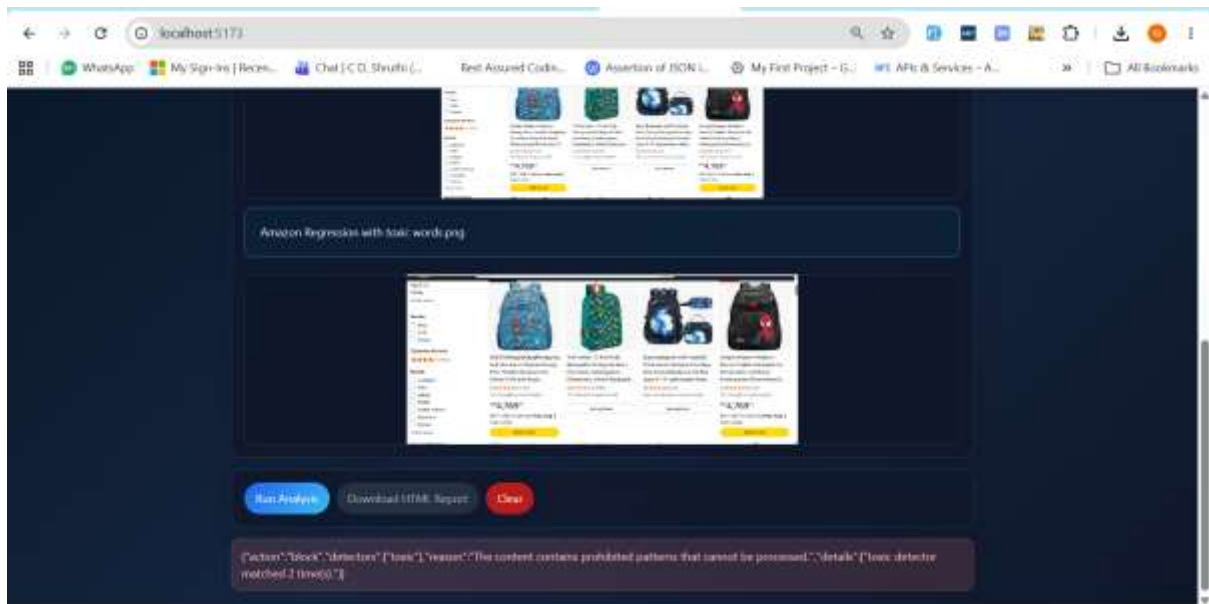


Amazon Regression with toxic words.png



On clicking Run Analysis, moderation api is called and the analysis is blocked:





Moderation enabled = false

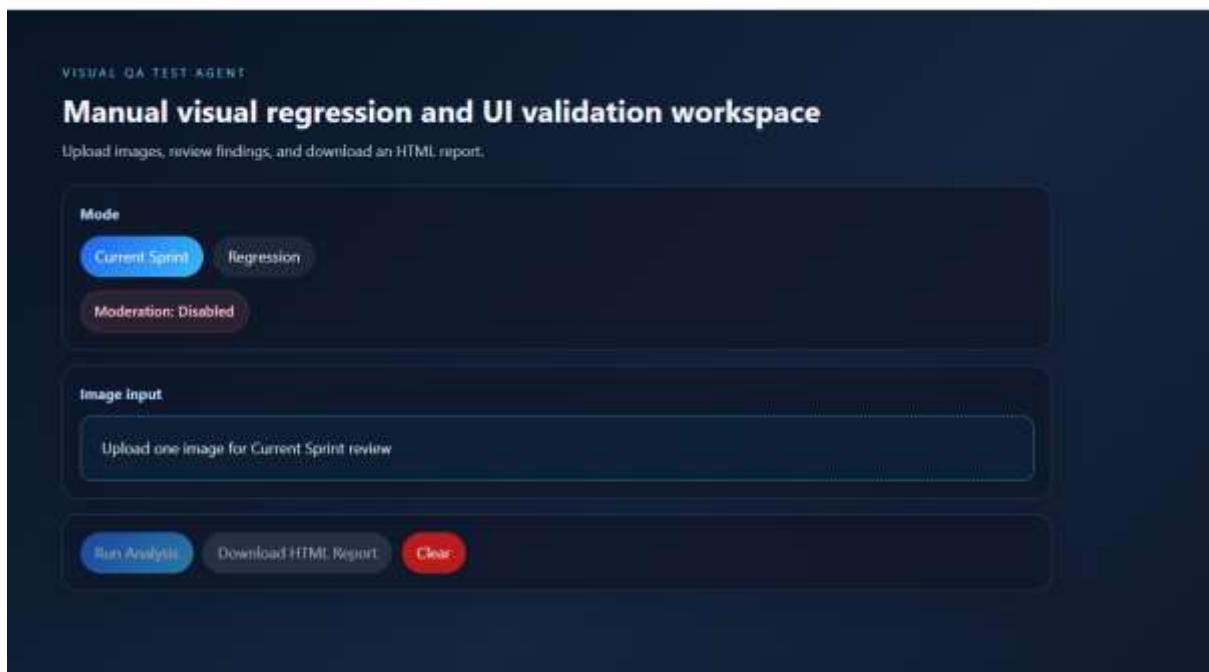


Image with toxic keyword is uploaded and no moderation call is made and the analysis is successful:



Amazon Regression with toxic words.png



Run Analysis

Download HTML Report

Clear

- Retake the screenshot if the image is blurry, cropped, or otherwise hard to inspect.
- Use the generated report as a handoff artifact for engineering or design follow-up.

Text difference analysis

Text comparison found a likely mismatch in the product title area between Amazon New feature.png and Amazon Regression with toxic words.png.

Detected change location: Product title mismatch

Baseline title text: Baseline title text in Amazon New feature.png: "Simple Modern Medium Disney Pixar Toddler Backpack for School Girls and Boys"

Baseline snippet: Simple Modern Medium Disney Pixar Toddler Backpack for School Girls and Boys

Current title text: Current title text in Amazon Regression with toxic words.png: "9864784tugh868ygjtuhgurg Simple Modern Medium Disney Pixar Toddler Backpack for School Girls and Boys"

Current snippet: 9864784tugh868ygjtuhgurg Simple Modern Medium Disney Pixar Toddler Backpack for School Girls and Boys

Likely impact: A corrupted or inconsistent product title can confuse shoppers, reduce trust, and degrade the perceived quality of the search results.

Confidence: high

- The baseline screenshot shows the expected product title while the current screenshot contains an unexpected character sequence inserted into the same title area.
- The difference is in the product listing headline, not a checkout button label or sidebar control.

Pixel diff

37.8% mismatch between the baseline and current images.

Mismatch exceeds the configured threshold.

Baseline image is 1915x1009, current image is 1915x829. Pixel diff was calculated over the overlapping 1915x829 region.

The current image appears blurrier than the baseline, which is consistent with the visible softening or loss of edge detail in the regression capture.

► View pixel diff preview